

Total No. of Questions : 8]

SEAT No. :

PC-2370

[Total No. of Pages : 2

[6354]-487

B.E. (Computer Engineering)
BLOCKCHAIN TECHNOLOGY
(2019 Pattern) (Semester - VII) (410243)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Solve questions Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Discuss in brief Bitcoin and Ethereum cryptocurrencies. **[8]**

b) Explain following : **[9]**

- i) Proof of work
- ii) Proof of stake and
- iii) Proof of activity

OR

Q2) a) List and explain types of Blockchain. **[8]**

b) Explain Byzantine general problem scenario. Explain the problem and its probable consequences. **[9]**

Q3) a) Differentiate between Coinbase and Binance. **[9]**

b) What is Metamask? Discuss any one application of Metamask. **[8]**

OR

Q4) a) List and explain types of crypto wallet **[8]**

b) Write a note on Bitcoin. **[9]**

P.T.O.

Q5) a) What is Swarm, and how does it address the need for decentralized storage in the Ethereum network? [9]

b) What is Ethereum? Define smart contracts and give an example of a real-world scenario where a smart contract could be beneficial. [9]

OR

Q6) a) State and explain the various components of Ethereum. Explain the various types of Ethereum Networks. [9]

b) Explain Decentralized Messaging Platform - Whisper and its purpose in the Ethereum ecosystem. [9]

Q7) a) With the help of neat diagram explain how blockchain can be used for any supply chain application. [9]

b) Explain the use of blockchain technology in the government sector. 'Which are the different blockchain based applications proposed for it? [9]

OR

Q8) a) Explain how blockchain and IOT can work together. [9]

b) List out the applications of blockchain Technology in different areas. Explain any 2 in detail. [9]
